Different observers slice up the loaf at different angles, depending on details of their relative motion. In the examples encountered previously, we did not take account of expanding space and, instead, imagined that the fabric of the cosmos was fixed and unchanging over time. We can now refine those examples by including cosmological evolution.

To do so, we will take the perspective of observers who are at rest with respect to space—that is, observers whose only motion arises from cosmic expansion, just like the Lincolns glued to the balloon. Again, even though they are moving relative to one another, there is symmetry among all such observers—their watches all agree—and so they slice up the spacetime loaf in exactly the same way. Only relative motion in excess of that coming from spatial expansion, only relative motion *through* space as opposed to motion *from* swelling space, would result in their watches falling out of synch and their slices of the spacetime loaf being at different angles. We also need to specify the shape of space, and for purposes of comparison we will consider some of the possibilities discussed above.

The easiest example to draw is the flat and finite shape, the video game shape. In Figure 8.7a, we show one slice in such a universe, a schematic image you should think of as representing all of space right *now.* For simplicity, imagine that our galaxy, the Milky Way, is in the middle of the figure, but bear in mind that no location is in any way special compared with any other. Even the edges are illusory. The top side is not a place where space ends, since you can pass through and reappear at the bottom; similarly, the left side is not a place where space ends, since you can pass through and reappear on the right side. To accommodate astronomical observations, each side should extend at least 14 billion light-years (about 85 billion trillion miles) from its midpoint, but each could be much longer.

Note that right now we can’t literally see the stars and galaxies as drawn on this *now* slice since, as we discussed in Chapter 5, it takes time for the light emitted by any object right now to reach us. Instead, the light we see when we look up on a clear, dark night was emitted long ago—millions and even billions of years ago—and only now has completed the long journey to earth, entered our telescopes, and allowed us to marvel at the wonders of deep space. Since space is expanding, eons ago,